Guneesh Vats

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Machine Learning Engineer building production ML systems across LLMs, retrieval, multimodal learning, and speech. Experience spans model development, evaluation, inference optimization, and ML-backed services, supported by a strong research background.

Education

International Institute of Information Technology (IIIT), Hyderabad *Aug 2020 – Jun 2025*
B.Tech and M.S. (Research), Electronics and Communication Engineering (ECE) MS GPA: **8.70/10** | B.Tech GPA: **8.00/10**
MS Thesis: Change-Point Detection via Structured Projections: From Channel Codes to Embeddings
Honors: Dean's List (Top 1% – Monsoon '21, Spring '23) | 7th globally, IEEE ICASSP Grand Challenge 2025 | Flipkart Grid 2023 Finalist

Experience

Machine Learning Engineer – **Coschool AI** *Jun 2026 – Present | Hyderabad*

- **Question-level Multimodal Evaluation:** Built a handwritten-answer evaluation system with rubric-guided VLM scoring, achieving **95% accuracy**; direct-image evaluation reduced inference latency and cost by 35%.
- Cut per-request LLM input-token cost by **70%** ($0.0125 → $0.0038; 36% lower total cost) via **prompt prefix caching**, raising cache hits from 53% to 87% with cyclic cache keys while preserving grading stability.
- Improved RAG retrieval efficiency by adding a **cross-encoder re-ranker** over cosine-retrieved candidates and calibrating score-based filtering, reducing prompt tokens by **~40%** while maintaining grading quality.
- **Worksheet Segmentation Service:** Fine-tuned YOLOv11 (nano & small) for label/separator detection (**mAP50 0.975**) and deployed a Kubernetes-hosted FastAPI service supporting VLM question matching, cross-page answer stitching, and separator recovery.
- **Worksheet-level Batched Grading:** Evaluated every question in an answer script against its rubric in a single VLM call; benchmarked 8 GPT and 5 Claude models, **reducing input tokens by ~75% and evaluation cost by ~47%** without mis-mapped answers.
- Devised a Gaussian Mixture Model–based classifier to detect on- and off-topic queries in chatbot conversations with 95% accuracy.

Applied Research Scientist (ARS) – **ServiceNow** *Jan 2026 – Jun 2026 | Hyderabad*

- **REDACT – Multilingual PII Benchmark:** Built an EMNLP 2026 benchmark with **13,427 records and 324K annotations** across 51 entity types, 25 languages, and 9 scripts for controlled failure analysis.
- Engineered a **7-stage LLM data pipeline** for controlled generation, deterministic span alignment, verification, repair, semantic dedup, and release auditing; incorporated disclosure form and sensitivity metadata to support fine-grained evaluation.
- Evaluated **Presidio, GLiNER, OpenAI Privacy Filter & frontier LLMs** using exact, partial, and fuzzy span matching, exposing failures across language, entity type, disclosure form, and sensitivity tier.
- **ITSM Agent Benchmark:** Built a **13-stage synthetic-data pipeline** converting patterns from enterprise ITSM incidents into **1,000 synthetic incidents with 43 fields**, with normalization, clustering, stratified sampling, validation, repair, and re-validation.
- Architected a **3-tier ITSM evaluation framework** culminating in a simulated ServiceNow environment with **4 workflows, 7 ReAct agents, and 20 workflow-facing tools** for triage, retrieval, routing, resolution planning, action execution, and post-incident review.
- Built a **2,280-article knowledge base** with hard negatives and abstention cases; designed **75+ metrics across 7 dimensions** covering correctness, tool efficiency, workflow completion, information use, and safety.

Founding Machine Learning Engineer – **Alonzo AI** *Sep 2024 – Dec 2025 | Hyderabad*

- **Sports Intelligence CMS:** Architected a multimodal platform using fine-tuned YOLO and AWS Textract to digitize legacy record books, reconcile statistics against production schemas, and support source-linked review, correction, approval, and publishing.
- Developed a rare-event discovery and publishing layer over the structured sports database, combining rarity attributes, historical context, and record-level metadata to surface noteworthy performances and support direct publishing to social channels.
- **Conversational Sports Analytics:** Built a schema-aware engine that extracts entities, metrics, filters, and time constraints, maps them to supported query types, and populates validated SQL/MongoDB templates for execution in the CMS.
- Built a **schema-aware retrieval layer** with SBERT embeddings indexed in Milvus, combining metadata filtering, cross-encoder reranking, and score-based context pruning to retrieve relevant schemas and fields; **reduced query latency by 40% and LLM API cost by 30%**.
- Built an **automated evaluation and regression framework** validating information extraction, schema mapping, generated queries, and execution results across client schemas, achieving **95% extraction accuracy and 98% database-query accuracy**; used pytest for functional regression and JMeter/Locust for latency and load testing.

ML Engineer Intern (Speech) – **Adalat AI** *Apr 2024 – Jul 2024 | Remote*

- Reduced court **ASR word-error rate to 21.4% (18pp improvement)** on a 30K-minute multi-speaker dataset by fine-tuning Whisper via **LoRA/PEFT**, with PyAnnote diarization and log-mel alignment.
- Built a **real-time speech-recognition inference and model-serving backend** using FastAPI and NVIDIA Riva, achieving **~100 ms latency** through GPU caching, batching, token prefetching, and speaker-aware preprocessing.

Recent Publications

Complete list on Google Scholar.

- **REDACT: A Systematically Controlled Multilingual Benchmark for Personal Information Detection** – *EMNLP 2026, Published*
  PII benchmark spanning 25 languages, 9 scripts, 51 entity types, and 324K annotations.
- **Neural-Network Based Change-Point Detection for Binary Linear Block Codes** – *IEEE WCNC 2026, Published*
  Detects code transitions using structured projections and neural refinement.
- **Detecting Convolutional Codes via a Markovian Statistic** – *IEEE WCNC 2026, Published*
  Introduces a Markovian statistic for blind convolutional-code detection.
- **Decoding Emotion: Speech Perception Patterns in Individuals with Self-reported Depression** – *IEEE O-COCOSDA, 2024, Published*
  Analyzes depression-linked differences in affective speech perception.

Technical Skills

- **Languages & ML:** Python, C/C++, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
- **LLMs & Retrieval:** LLM Fine-Tuning (LoRA/PEFT), LLM Evaluation, Prompt version control, RAG, Milvus, Pinecone, ChromaDB
- **Multimodal & Speech:** VLMs, YOLO, OCR/Document AI, Whisper, PyAnnote, NVIDIA Riva & NeMo
- **ML Systems & Serving:** FastAPI, Docker, Kubernetes, CUDA, ONNX, AWS (EC2, S3, SageMaker), GitHub Actions, Kubernetes, Flask
- **Data, Observability & Testing:** SQL, PostgreSQL, MongoDB, Redis, Neo4j, Prometheus, Grafana, pytest, JMeter, Locust, A/B Testing